Chaos and Fractals: Patterns from Simple Rules

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November 12, 2026

Abstract

This project explores how simple repeated rules can produce intricate patterns, connecting the study of chaos with the geometry of fractals. Chaos concerns systems whose behavior can become difficult to predict because small differences in their starting conditions grow over time, while fractals describe shapes and sets with detail at increasingly small scales. These ideas often meet, but fractal structure does not by itself imply chaotic behavior. Using Chapter 11 of Strogatz's *Nonlinear Dynamics and Chaos* as a foundation, we will consider how mathematicians describe the size and complexity of such patterns through countability, uncountability, and fractal dimension. The Cantor set provides a central example: removing intervals repeatedly leaves an uncountable collection of points with zero total length. Box dimension offers another way to understand its size by tracking how many small boxes are needed to cover it as the scale becomes finer.

Two examples will connect these abstract ideas to the emergence of recognizable patterns: the chaos game and the Devil's staircase. In the familiar chaos game, repeatedly moving partway toward a randomly chosen vertex of a triangle produces the Sierpiński triangle, showing how random choices can reveal an organized geometric structure. Barnsley and Vince's *The chaos game on a general iterated function system* explains why this approach can recover an attracting set under broad conditions, extending beyond the usual setting in which every rule shrinks distances. Per Bak's *The Devil's staircase* examines a different source of intricate organization: systems with competing rhythms that synchronize over ranges of conditions, creating a hierarchy of plateaus separated by a Cantor-like set. Together, these examples frame the project around a shared question: how does repeated evolution produce structure across scales? By comparing a fractal generated by random iteration with fractal organization among the conditions for synchronization, we will use visual examples and accessible geometric ideas to connect unpredictability, order, and the different meanings of mathematical size.

1 Random iteration: the circle attractor

To reproduce Example 1 of Barnsley and Vince [2], we take the state space to be the unit circle and at each step independently choose either the identity or a rotation, each with probability $1/2$. We use the irrational rotation angle $2\pi(\sqrt{5}-1)/2$ and start at $(1,0)$. Both maps preserve distances, so this example tests the chaos game beyond the familiar contracting maps that generate the Sierpiński triangle.

After n steps the possible rotation counts are $0, 1, \ldots, n$. Irrational rotation makes the resulting deterministic sets dense in the limit, and the random run visits every successive rotation count almost surely because it keeps choosing the rotation infinitely often. This illustrates the conclusion of the paper's more general theorem in a setting where the attractor is known.

For a finite set of sampled points, let g_N denote the largest gap between successive distinct phases, measured in turns and including the wraparound gap. The point farthest from the sample is

at the midpoint of that gap, so the Euclidean Hausdorff distance to the circle is

$$d_H = 2\sin(\pi g_N/2).$$

The experiment uses a fixed random seed and measures cumulative coverage at powers of two. The error drops from 1.129 after 8 iterates to 0.00230 after 4096. Figure 1 reproduces the qualitative finding that a single random orbit draws the circle and adds a numerical measure of how well the sample covers it. The finite prefix calculation illustrates coverage; the almost sure statement about infinite tails comes from the argument above and the theorem, rather than the plot.

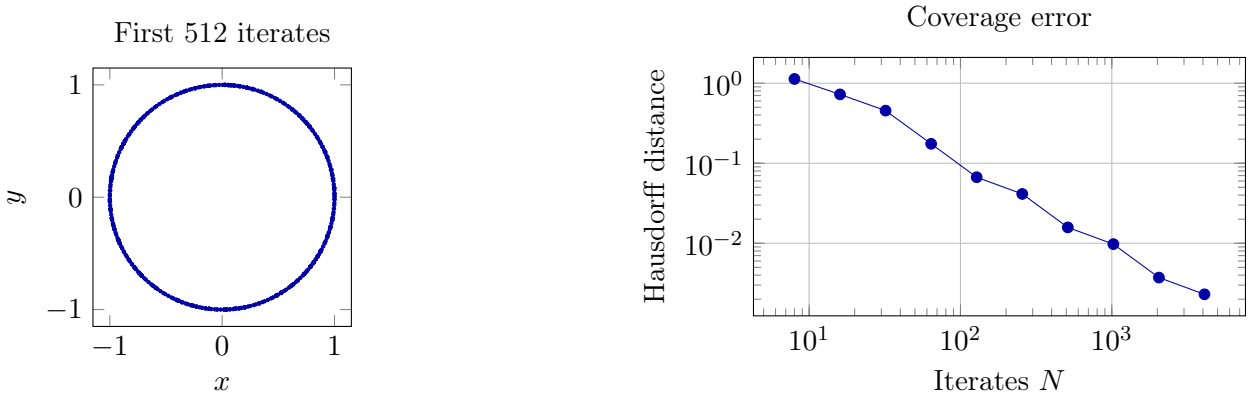


Figure 1: The noncontractive chaos game on the unit circle. Repeated points do not improve coverage; newly visited phases close the gaps.

2 Frequency locking: the critical Devil's staircase

To reproduce the winding number staircase in Figure 5(c) of Bak [3], we iterate the lift of the sine circle map,

$$x_{n+1} = x_n + \Omega + \frac{K}{2\pi}\sin(2\pi x_n).$$

The bias Ω is the uncoupled advance per step, and K sets the strength of the periodic term. At the critical value $K = 1$, we estimate the winding number by averaging the unwrapped increments after a transient. The implementation keeps the phase modulo one for evaluating the sine but accumulates the full increments. We use 3001 equally spaced values of Ω on $[0, 1]$, discard 4096 steps and average the next 16384, starting all runs at phase 0.123.

Figure 2 shows broad plateaus at 0, 1/2 and 1, with smaller steps including 1/3 and 2/3. For example, the computed winding number is within 2×10^{-5} of 1/2 at grid points from 0.463333 to 0.536667. This is a grid-based estimate, not a determination of the exact plateau endpoints. The maximum change across the grid when doubling both the transient and averaging windows from half their final values was 5.51×10^{-5}. As a separate check, a fixed point of the lift satisfies $\Omega = -\sin(2\pi x)/(2\pi)$ at $K = 1$. Thus the zero-winding plateau on $[0, 1]$ extends to $1/(2\pi) \approx 0.159155$, consistent with the last detected grid point at 0.159000. The finite grid resolves only some of the plateaus and cannot establish that the full staircase occupies full measure.

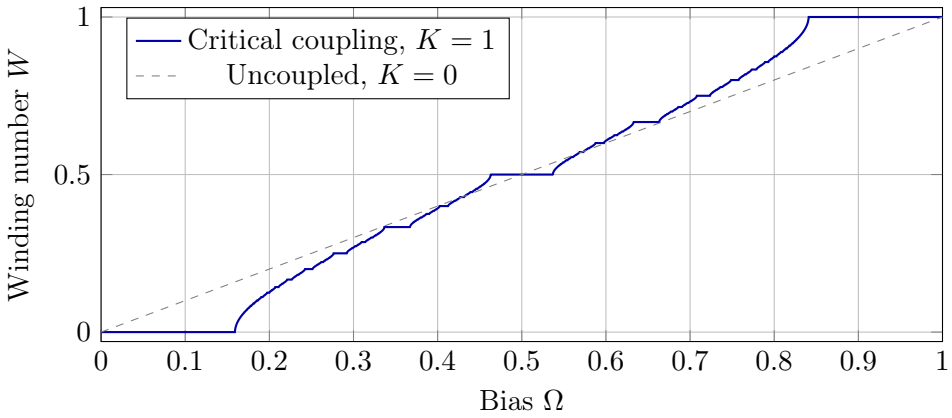


Figure 2: Numerical reproduction of Bak's critical winding number staircase. Horizontal segments indicate that the same average rotation persists over a range of biases.

3 The connection through fractal dimension

Strogatz [1, Chapter 11] distinguishes counting points, measuring length, and counting boxes as the observation scale shrinks. For instance, the middle-thirds Cantor set has zero length and uncountably many points, yet at scale 3^{-m} it takes on the order of 2^m boxes to cover it, giving box dimension $\log 2/\log 3$. These are three different ways to describe the same set.

The circle experiment concerns the geometry of an attractor in state space. Its orbit has only countably many points, yet its closure is the entire circle, with box dimension one. The staircase experiment concerns geometry in parameter space: its fractal set lies on the bias axis, where the open locking intervals have been removed. Bak estimates its dimension near 0.87 by measuring how much of the bias axis remains uncovered after accounting for plateaus wider than a scale r and dividing the remaining length by r. This plays a role analogous to Strogatz's box counting because both methods study how a count changes with resolution. We reproduce Bak's staircase, not the separate dimension estimate, which would require resolving many more widths across several scales. The shared lesson is that a finite orbit or a finite parameter scan shows structure at a chosen resolution; the limiting set explains what those observations are approaching.

References

[1] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. 2nd ed., CRC Press, 2018. Chapter 11, "Fractals."

[2] Michael F. Barnsley and Andrew Vince. "The chaos game on a general iterated function system." *Ergodic Theory and Dynamical Systems*, 31(4), 1073–1079, 2011. Published online in 2010. DOI: 10.1017/S0143385710000428.

[3] Per Bak. "The Devil's staircase." *Physics Today*, 39(12), 38–45, 1986.